

Rule Interpretation Categories Guide

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This guide explains the eight interpretation categories used by the Rule Panel.

These categories are not generic chat labels and they are not new clustering or

outlier-detection algorithms. Their purpose is to translate decision-tree

surrogate rules into concrete label/refinement guidance: what the user should

label next, why those labels are informative, and how the results could affect

merge, split, new-cluster, boundary, anomaly, or relabel decisions.

1. System Role

The current responsibility boundary is:

- SSDBCODI owns current cluster assignments and outlier flags.
- The decision tree only converts SSDBCODI outputs into readable surrogate rules. It does not recluster data and does not redetect outliers.
- Rule interpretation converts those rules into label/refinement guidance.
- DeepSeek or the mock interpreter must not mutate cluster, outlier, selection, or labeling state.

The interpretation layer answers questions such as:

- Which points should be labeled first?
- Which rule relation needs human confirmation?
- Does the evidence suggest merge review, split review, new-cluster review, or only additional labeling?
- Which anomaly or exception points need confirmation?
- Are the current rules strong enough to guide refinement?

2. Output Structure

Each RuleInterpretation should include:

- `categories`: one or more category ids.
- `category_explanation`: a short explanation of what the category examines.
- `recommendation`: the next action the user should take.
- `summary`: a concise but detailed explanation of why the recommendation

matters.

- `label_targets`: the points the user should label now, plus the label question those points are meant to answer.
- `suspicion_reasons`: why those points are suspicious or strategically useful,
- `point_label_guidance`: how to label the selected points, including label

options, a concrete checklist, and the decision impact of likely outcomes.

This is the main LLM-authored point-level guidance.

- `decision_rationale`: strategic reasoning that connects rule evidence to a

labeling or refinement decision. This is where the LLM adds value beyond raw

metrics.

- `label_outcomes`: possible user-label outcomes and what each outcome would

imply.

- `quantitative_findings`: auditable numbers such as support, coverage, purity,

Jaccard overlap, exception count, and rule confidence score.

- `suggested_label_actions`: executable labeling actions. Each action should

include reason, hypothesis, why_this_action, expected_outcomes, and risk_note.

- `evidence`: traceable rule ids, feature thresholds, target ids, and point ids.

- `warnings`: limitations such as no sample-level overlap or no exception

points.

2.1 What the LLM Adds

The system computes quantitative metrics in code so numeric claims remain

auditable. The LLM should not invent support, overlap, thresholds, or point ids.

The LLM's real value is synthesis:

- turn rule overlap, boundary gaps, and exception points into testable hypotheses;
- explain why the selected points are more informative than random points;
- combine point-level raw feature values, threshold margins, current cluster assignment, and outlier status into concrete labeling guidance;
- describe what different label outcomes would imply;
- surface uncertainty and risk;
- convert raw feature thresholds into a user-facing labeling checklist.

A weak interpretation says only: "Jaccard overlap is 0.20."

A useful interpretation says: "The two rules have Jaccard overlap 0.20. Label

wine_014 and wine_021 first because they test both the overlap and the boundary

semantics. If those points receive one semantic label, merge or shared-boundary

review becomes plausible. If the labels are mixed, the region is more likely a

boundary ambiguity or a new-cluster candidate."

When the system recommends inspecting raw feature thresholds, it is not saying

that the raw feature is wrong. The actual doubt is that a decision-tree threshold only proves the surrogate can reproduce current SSDBCODI output; it

does not prove that the threshold matches the user's semantic labels. The user

should therefore compare candidate points' raw feature values and threshold

margins against their human label judgment.

3. Common Metrics

Support

support_count is the number of points matched by a rule. High support means

the rule covers a larger region. Low support can still be useful, especially

for anomaly or exception inspection, but it should not define an entire cluster

by itself.

Coverage

coverage is the fraction of the target class covered by the rule. High

coverage means the rule is representative of its target. Low coverage means the

target may require multiple rule regions, which can indicate split or new-cluster review.

Purity

purity is the fraction of matched points that already belong to the rule

target under the current SSDBCODI output. High purity means the rule is clean

with respect to SSDBCODI. Low purity indicates mixed membership and should

trigger label review.

Rule Confidence Score

rule_confidence_score = purity * coverage. This is not a probability. It is a

ranking signal. A high score identifies stable rule regions; a low score identifies rules that need audit before refinement.

Exception Count and Exception Rate

Exception points are matched by a rule but disagree with that rule's target.

They are often the most valuable points to label because they expose direct

conflict between the surrogate rule and the current SSDBCODI assignment.

Pair Intersection Count

intersection_count is the number of matched points shared by two rules.

It is

the core evidence for overlap-based merge review. If two rules with different

targets share many points, those shared points should be labeled first.

Jaccard Overlap

jaccard_overlap = intersection / union. It measures similarity between two

rule match sets. If sample-level overlap is zero, the system must not recommend

a merge from rule cards alone.

Overlap Share A / B

overlap_share_a and overlap_share_b describe how much of each rule is covered by the pair intersection. A small rule mostly covered by a larger rule

may be a subregion, a boundary slice, or an anomaly pocket.

Boundary Gap

boundary_gap describes whether two rules are adjacent, overlapping, or separated along shared raw features. Small or zero gaps are useful for boundary

review.

4. Category: Label Priority

Meaning

Label Priority answers: "Which points or rule regions should the user label first?"

When To Use

Use this category when the user needs a global next step rather than a specialized explanation. It is the best default overview category.

Evidence Used

- highest-priority rule pair;
- label candidate groups;
- support, coverage, and purity;
- exception count;
- overlap or boundary relation;
- representative matched points.

Expected Output

The output should state:

- which point ids should be labeled first;
- which rule ids they come from;
- why those labels are more useful than random labels;
- what refinement decision the labels could influence.

How The User Should Use It

Treat Label Priority as an active-learning suggestion. It helps the user spend labeling effort where it will most reduce uncertainty.

Example

If the strongest rule covers 43 points with purity 1.00 and coverage 1.00, the

system may suggest labeling representative points from that rule to verify it

as a stable reference region. If another rule pair has cross-cluster overlap,

Label Priority should instead prioritize the overlap points.

Caution

Label Priority does not change the model. It only chooses where to inspect

first.

5. Category: Boundary Review

Meaning

Boundary Review answers: "Does a rule boundary correspond to a real semantic boundary?"

When To Use

Use it when two cluster rules are adjacent in raw feature space, when their

boundary gap is small, or when different target regions are close.

Evidence Used

- shared raw feature names;
- threshold intervals;
- boundary gaps;
- pair relations such as adjacent_cluster_boundary;
- support, coverage, and purity on both sides;
- candidate point ids near the boundary.

Expected Output

The output should explain:

- which two rules define the boundary;
- which raw feature threshold forms the boundary;
- which points should be labeled on each side;
- what same-label versus different-label outcomes would imply.

How The User Should Use It

Label paired points from both sides of the boundary. A boundary cannot be

confirmed by labeling only one side.

Example

If one rule is proline ≤ 484 and the adjacent rule is proline > 484 and

proline ≤ 645 , the system may suggest labeling representative points from both sides of proline = 484.

Caution

Boundary Review is not a direct merge recommendation. It only says the boundary should be tested.

6. Category: Overlap Merge Signal

Meaning

Overlap Merge Signal answers: "Do overlapping rules suggest that two targets

may share one semantic region?"

When To Use

Use it when two rules have sample-level overlap, especially if they belong to

different targets.

Evidence Used

- pair intersection count;
- Jaccard overlap;
- overlap share for each rule;
- shared point ids;
- rule target kinds and target ids;
- shared features and thresholds.

Expected Output

The output should state:

- how many points are shared;
- the Jaccard overlap;
- the overlap share for both rules;
- which shared points should be labeled;
- what same-label and mixed-label outcomes would imply.

How The User Should Use It

Label overlap points first. If they receive the same semantic label, merge or

shared-boundary review becomes plausible. If labels are mixed, keep targets

separate or investigate a new cluster.

Example

If rule_cluster_2 and rule_cluster_3 share 12 points with Jaccard overlap

0.30, those 12 shared points should be labeled before proposing merge.

Caution

If intersection_count = 0, the system must not recommend merge from rule cards alone. It should fall back to Boundary Review or Label Priority.

7. Category: Split Or New Cluster Signal

Meaning

Split Or New Cluster Signal answers: "Does an existing target appear to contain

multiple semantic subregions?"

When To Use

Use it when the same target requires multiple disjoint rules, when coverage is

low but purity is high, or when separated regions appear inside one cluster.

Evidence Used

- same-target disjoint regions;
- low coverage with high purity;
- multiple rules for one target;
- boundary gaps and shared features;
- exception groups;
- candidate point groups.

Expected Output

The output should identify:

- which target may contain separate regions;
- which rule ids represent those regions;
- support, coverage, and purity for each region;
- which representative points should be labeled;
- what label outcomes would support split or new-cluster review.

How The User Should Use It

Label across multiple regions. A split decision requires evidence that the

regions receive different human labels.

Example

If one cluster needs three disjoint rules with different feature paths,
label
representatives from each rule. Different labels support split; a label
not
matching any current cluster can support new-cluster review.

Caution

Split and new-cluster actions are expensive. They should remain
hypotheses until
human labels support them.

8. Category: Anomaly Label Review

Meaning

Anomaly Label Review answers: "Are current outlier-rule points true anomalies

or normal members near a boundary?"

When To Use

Use it when anomaly rules exist, when anomaly support is small, when coverage is

low, or when anomaly regions are near cluster rules.

Evidence Used

- anomaly rule support;
- anomaly rule coverage;
- anomaly point ids;
- cluster-anomaly relation;
- outlier score range;
- matched anomaly points.

Expected Output

The output should state:

- which anomaly points need confirmation;
- which anomaly rule matched them;
- the rule's support, coverage, and purity;
- what true-anomaly versus normal-member labels would imply.

How The User Should Use It

Label candidate points as true anomaly, normal member, or cluster member. These

labels help decide whether the outlier flag should be trusted.

Example

If an anomaly rule matches only one point with coverage 0.06, the system should

recommend direct human confirmation rather than treating the rule as a stable anomaly pattern.

Caution

This category does not automatically remove outlier flags. It identifies outlier candidates that need user confirmation.

9. Category: Exception Relabel Review

Meaning

Exception Relabel Review answers: "Which rule exceptions should be relabeled or used to fix boundaries?"

When To Use

Use it when rules expose `exception_point_ids`, low purity, or mixed target membership.

Evidence Used

- exception point ids;
- exception count;
- exception rate;
- rule purity;
- raw feature thresholds for the exception rule;
- whether exception points cluster into a small region.

Expected Output

The output should explain:

- which exception points need labels;
- which rule they contradict;
- exception count and exception rate;
- whether repeated exception labels would support relabel, boundary fix, or new cluster review.

How The User Should Use It

Label exception points before ordinary matched points. They are direct evidence of disagreement between current assignments and the surrogate explanation.

Example

If rule_cluster_2 matches 20 points and 4 of them currently belong to cluster_3, those 4 exception points should be prioritized.

Caution

Some configurations may produce no exception points. In that case the system

should warn, not invent relabel candidates.

10. Category: Feature Label Strategy

Meaning

Feature Label Strategy answers: "Which raw feature thresholds should guide

manual labeling?"

When To Use

Use it when certain raw features appear repeatedly in rules or when a threshold

is central to cluster or anomaly separation.

Evidence Used

- feature usage counts;
- rule conditions;
- repeated thresholds;
- strongest rule paths;
- support, coverage, and purity;
- candidate points.

Expected Output

The output should state:

- the most important raw features;
- repeated thresholds;
- which point ids should be labeled while checking those feature values;
- why projected x/y position is insufficient.

How The User Should Use It

When labeling candidate points, inspect raw features such as proline, magnesium, or flavanoids. Do not rely only on the 2D scatterplot location.

Example

If proline appears 9 times and magnesium appears 2 times in the rules,
the
system should tell the user to treat those features as the labeling
checklist.

Caution

Feature importance here is surrogate importance, not causal importance.

11. Category: Rule Confidence Audit

Meaning

Rule Confidence Audit answers: "Are the rules reliable enough to guide refinement?"

When To Use

Use it before merge, split, new-cluster, or anomaly decisions. It is also useful

for broad, narrow, low-purity, low-coverage, or high-exception rules.

Evidence Used

- support;
- coverage;
- purity;
- rule confidence score;
- exception count and exception rate;
- quality warnings;
- condition count;
- matched point previews.

Expected Output

The output should identify:

- stable rules;
- weak rules;
- rules that need more labels;
- rules that should not yet drive refinement.

How The User Should Use It

Treat it as a safety check. If rule quality is weak, gather labels before

changing cluster or anomaly state.

Example

A rule with purity 1.00 but coverage 0.06 is clean but local. It is useful for inspection, not for defining a whole cluster.

Caution

High confidence means consistency with SSDBCODI, not semantic correctness.

Human labels remain the source of semantic truth.

12. Choosing A Category

- Unsure where to start: use `label_priority`.
- Suspect a weak cluster boundary: use `boundary_review`.
- Two rules share matched points: use `overlap_merge_signal`.
- One cluster appears to contain separated regions: use `split_or_new_cluster_signal`.
- Inspecting outliers: use `anomaly_label_review`.
- Exception points exist: use `exception_relabel_review`.
- Need raw-feature guidance for labeling: use `feature_label_strategy`.
- Before acting on rules: use `rule_confidence_audit`.

13. Core Principles

1. Rule interpretation provides label guidance; it does not mutate state.
2. The decision tree is a surrogate rule extractor, not a new clustering algorithm.
3. Merge, split, and new-cluster decisions require human labels.
4. If sample-level overlap is zero, do not recommend merge directly.
5. Low-support or low-coverage rules can still be useful for local inspection.
6. Exception points are often more valuable than ordinary matched points.
7. Projection is helpful for viewing, but explanations should return to raw feature thresholds.